THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 9: **Life Processes in Plants** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A leaf from a plant kept in the dark and another from sunlight are tested with iodine. The one that turns blue-black is: 1

(a) the sunlight leaf **(b)** the dark leaf
(c) both **(d)** neither

2. A potted plant is covered with a polythene bag and kept in sunlight. After some time water droplets are seen inside the bag. This shows: 1

(a) respiration **(b)** photosynthesis **(c)** digestion **(d)** transpiration

3. Cuscuta has no chlorophyll. It survives by: 1

(a) absorbing from dead matter **(b)** taking food from its host
(c) eating insects **(d)** making food

4. In the variegated leaf, the white parts are tested with iodine. They: 1

(a) remain brown **(b)** turn red **(c)** turn blue-black **(d)** turn green

5. Which plant cell part controls the opening of stomata? 1

(a) guard cells **(b)** root hair **(c)** phloem **(d)** xylem

6. A mushroom growing on a log gets its food from: 1

(a) insects **(b)** air **(c)** sunlight **(d)** dead wood

7. A leaf covered partly with black paper is tested with iodine. The covered part: 1

(a) turns blue-black **(b)** turns red **(c)** remains brown **(d)** turns green

8. The root hairs increase the surface for absorption of: 1

(a) water and minerals **(b)** light
(c) oxygen only **(d)** food

9. A pitcher plant in nitrogen-poor soil survives by: 1

(a) sleeping **(b)** growing taller
(c) trapping insects **(d)** absorbing more water

10. If carbon dioxide is absorbed by potassium hydroxide in a closed jar and a leaf inside is tested: 1

(a) no starch is formed **(b)** more starch
(c) nothing **(d)** green colour

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are potted plants kept near a window grown healthier than those kept in the dark? 2

12. Why do plants have many stomata on the lower surface of the leaf? 2

13. Why are leguminous plants good for crop rotation? 2

14. Explain why a plant kept in the dark for a few days loses starch from the leaves. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Design an experiment to show that CO2 is needed for photosynthesis. 3

16. Explain how stomata open and close. 3

17. Describe how a student can show that oxygen is released in photosynthesis. 3

OR

Explain how the transport of water and food in plants differ. 3

18. Explain why the photosynthesis rate is high at noon and low at dawn. 3

OR

Explain why a field of pulses needs less fertiliser. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how a leaf is adapted for photosynthesis. 5

OR

Describe the experiment on transpiration with a polythene bag and the precautions. 5

20. Describe the experiment to show that light is necessary for photosynthesis, with the steps, observation and conclusion. 5

OR

Explain the relationship of lichen and Rhizobium as examples of symbiosis. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND